概览

答卷

提交列表

排名

R1-1 分数 2 作者 何钦铭 单位 浙江大学

To find the minimum spanning tree of graph G(V,E) by ***Prim's algorithm***, it is possible that the first collected edge is the longest one in E where |E|>2 and each edge's length is different.

◉ T ○ F

答案正确：2 分 创建提问

R1-2 分数 2 作者 陈越 单位 浙江大学

In the syntax tree of an infix expression $A \times B + C - D/E$, the left child of the node D is a thread and is pointing to the node $/$.

○ T ◉ F

答案正确：2 分 创建提问

R1-3 分数 2 作者 何钦铭 单位 浙江大学

If the length of each edge in an undirected graph G(V,E) is increased by 1, the shortest path between any pair of nodes v_i and v_j will keep unchanged.

○ T ◉ F

答案正确：2 分 创建提问

R1-4 分数 2 作者 干红华 单位 浙江大学

In the binary MAX heap built from the array { 25, 14, 28, 51, 27, 11, 33, 20, 39, 23 }, the index of 39 is 1 (note that the index starts from 0)

◉ T ○ F

答案正确：2 分 创建提问

R1-5 分数 2 作者 何钦铭 单位 浙江大学

Supposed that the time complexity of function B(n) is $\Theta(n)$. Then the time complexity of the following program is $\Theta(n^2)$.

```
for ( i=n; i>0; i=i/2)
    B(n);
```

○ T ◉ F

答案正确：2 分 创建提问

R1-6 分数 2 作者 杨子祺 单位 浙江大学

Given an array 36, 95, 32, 58, 26, 38, 79, 46, 31, after the heapsort's initial min heap building of the array, the level order sequence of heap is 26, 31, 32, 36, 95, 38, 79, 46, 58.

◉ T ○ F

答案正确：2 分 创建提问

R1-7 分数 2 作者 朱建科 单位 浙江大学

A collision in hashing means two elements with different keys share the same hash value.

◉ T ○ F

答案正确：2 分 创建提问

R1-8 分数 2 作者 干红华 单位 浙江大学

In the binary search tree built by inserting 3, 1, 2, 5, 4, 6, 7 into an empty tree, 2 and 6 are on the same level.

◉ T ○ F

答案正确：2 分 创建提问

R1-9 分数 2 作者 何钦铭 单位 浙江大学

If a queue is implemented by a circularly linked list, then the insertion and deletion operations can be performed with one pointer ***rear*** instead of ***rear*** and ***front***.

○ T ◉ F

答案错误：0 分 创建提问

R1-10 分数 2 作者 杨子祺 单位 浙江大学

When performing the quick sort, after the second run or more runs, you can get a sequence like {11, 4, 19, 37, 25, 42, 51, 45} (assuming the pivot can be chosen randomly).

○ T ◉ F

答案正确：2 分 创建提问

概览

答卷

提交列表

排名

R2-1 分数 3 作者 干红华 单位 浙江大学

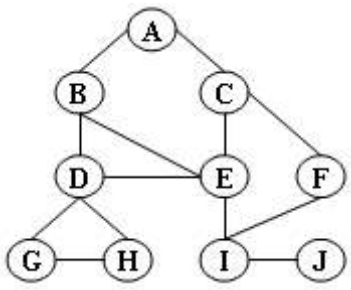
Given a binary search tree with its postorder traversal sequence { 1, 4, 13, 9, 21, 19, 31, 23, 17 }. If 17 is deleted from the tree, which one of the following statements is FALSE?

- A. It is possible that the preorder traversal sequence of the resulting tree may be { 19, 9, 4, 1, 13, 23, 21, 31 }
- B. It is possible that 13 and 21 may be on the same level
- C. It is possible that the preorder traversal sequence of the resulting tree may be { 21, 9, 4, 1, 13, 23, 19, 31 }
- D. It is possible that the resulting tree may have 3 leaves

答案正确：3 分 创建提问

R2-2 分数 3 作者 陈越 单位 浙江大学

How many biconnected components are there in the undirected graph below?



- A. 3
- B. 5
- C. 6
- D. 4

答案正确：3 分 创建提问

R2-3 分数 3 作者 陈越 单位 浙江大学

Which of the following algorithms is the most suitable one for finding a minimum spanning tree of a sparse graph?

- A. Prim
- B. Kruskal
- C. DFS
- D. Dijkstra

答案正确：3 分 创建提问

R2-4 分数 3 作者 陈越 单位 浙江大学

Given an array of integers { 7, 3, 6, 2, 5, 1, 4 }. Build a min-heap using the linear algorithm. Then which one is the list of nodes along a path from the root to a leaf node of the resulting heap?

- A. 1, 2, 7
- B. 1, 4, 7
- C. 1, 3, 5
- D. 1, 6, 2

答案正确：3 分 创建提问

R2-5 分数 3 作者 朱建科 单位 浙江大学

Given input {4371, 1323, 6173, 4199, 4344, 1000, 1999,1789} and a hash function $h(X) = X\%100$. If the collisions are solved by separate chaining, then the indices of the input numbers in the hash table are:

A. 71, 23, 73, 99, 44, 0, 99, 89

B. 71, 23, 73, 99, 44, 0, 1, 89

C. 71, 23, 73, 99, 44, 0, 0, 89

D. 71, 77, 73, 1, 44, 0, 0, 89

答案正确：3 分　创建提问

R2-6　分数 3　　作者 杨子祺　单位 浙江大学

Given an integer sequence { 31, 82, 34, 51, 22, 64, 73, 52 }. After the second run of the iterative merge sort in descending order, the integer sequence is

A. 82, 51, 34, 31, 73, 64, 52, 22

B. 82, 51, 34, 31, 52, 64, 73, 22

C. 82, 51, 34, 31, 64, 22, 52, 73

D. 82, 31, 34, 51, 73, 64, 52, 22

答案正确：3 分　创建提问

R2-7　分数 3　　作者 陈翔　单位 浙江大学

We represent a directed graph as an adjacency list:

```
v0: -> v5 -> v3 -> null
v1: -> null
v2: -> v1 -> null
v3: -> v4 -> v2 -> null
v4: -> v1 -> null
v5: -> v4 -> null
```

Which one below is not a topological order of the given graph?

A. 0 5 4 3 2 1

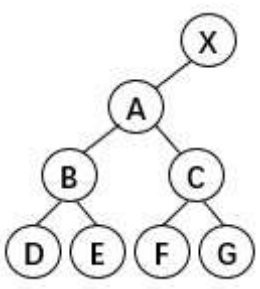
B. 0 5 3 4 2 1

C. 0 3 5 2 4 1

D. 0 3 2 5 4 1

答案正确：3 分　创建提问

R2-8　分数 3　　作者 陈越　单位 浙江大学



The above binary tree is the *first-child next-sibling* representation of the tree T. Then about the original tree T, which one of the following statements is **false**?

A. X must be the root and has three children

B. A has two children

C. C has two children

D. B has one child

答案正确：3 分　创建提问

R2-9　分数 3　　作者 何钦铭　单位 浙江大学

To convert an infix expression to a post-fix expression with a stack, which one of the following infix expressions has the ***maximum number of elements*** in stack during the conversion?

A. 2*(3*4+5)+6 B. 2*(3+4)*5+6 C. 2+(3*4+5)*6 D. 2*(3+4*5)+6

答案正确：3 分 创建提问

R2-10 分数 3 作者 朱建科 单位 浙江大学

Insert $\{18, 23, 17, 26, 31, 33, 4, 39\}$ one by one into a hash table of size 13 with the hash function $H(x) = x\%13$, and quadratic probing is used to resolve collisions. What is the loading density when the first collision occurs?

A. 0.62 B. 0.54 C. 0.31 D. 0.63

答案正确：3 分 创建提问

R2-11 分数 3 作者 何钦铭 单位 浙江大学

If a queue is implemented by a circular array of size 10, and the queue empty condition is ***front***==***rear***. When ***front***==3, and ***rear***==1, how many elements are there in the queue?

A. 9 B. uncertain C. 8 D. 2

答案正确：3 分 创建提问

R2-12 分数 3 作者 干红华 单位 浙江大学

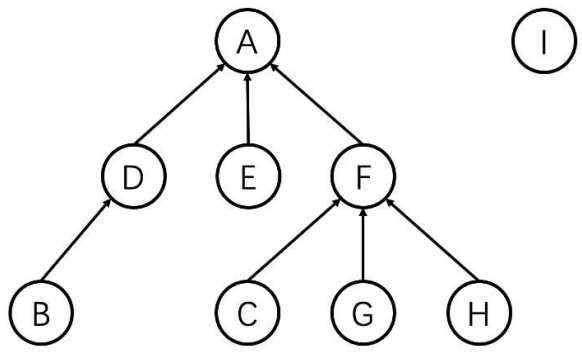
When inserting a new key K into a binary search tree T with 2024 nodes, the worst-case number of comparisons between K and the keys already in T is in the range of:

A. [10, 2023] B. [11, 2023] C. [11, 2024] D. [10, 2024]

答案正确：3 分 创建提问

R2-13 分数 3 作者 郑友怡 单位 浙江大学

The following is a forest formed after some number of UNIONs and FINDs, starting with the disjoint sets A, B, C, D, E, F, G, H, and I. Both union-by-size and path compression were used. Which routine can result in the following forest? Note that the UNION operation contains FIND, i.e., UNION(M, N) equals UNION(FIND(M), FIND(N)).



A. UNION(F, G), UNION(F, C), UNION(F, H), UNION(D, B), UNION(E, D), UNION(A, D), UNION(A, G)

B. UNION(D, B), UNION(A, D), UNION(A, E), UNION(F, C), UNION(F, G), UNION(F, H), UNION(A, F)

C. UNION(F, C), UNION(F, G), UNION(F, H), UNION(A, E), UNION(D, B), UNION(A, B), UNION(B, C)

D. UNION(D, B), UNION(F, C), UNION(A, E), UNION(G, H), UNION(A, B), UNION(F, H), UNION(E, H)

答案错误：0 分　创建提问

R2-14　分数 3　　作者 干红华　单位 浙江大学

Given a binary search tree with its postorder traversal sequence { 2, 1, 4, 9, 8, 3 }, the possible insertion order in which the given tree can be initialized is:

- A. 3, 8, 4, 9, 2, 1
- B. 3, 1, 8, 9, 2, 4
- C. 3, 1, 2, 4, 8, 9
- D. 3, 8, 2, 9, 1, 4

答案正确：3 分　创建提问

R2-15　分数 3　　作者 陈翔　单位 浙江大学

We represent an undirected graph as an upper triangular matrix (storing only the entries ***above*** the diagonal of the adjacency matrix), whose rows are then concatenated into a 1D-array: `[1 0 1 0 0 1 0 0 0 1 0 0 0 0 1]`. How many connected components are in the graph?

- A. 3 {{0, 1, 2}, {3}, {4, 5}}
- B. 2 {{0, 1, 2, 3}, {4, 5}}
- C. 4 {{0, 1, 2}, {3}, {4}, {5}}
- D. 1 {{0, 1, 2, 3, 4, 5}}

答案正确：3 分　创建提问

R2-16　分数 3　　作者 杨子祺　单位 浙江大学

Given the following sequence of numbers: { 52, 37, 63, 45, 71, 58, 42, 89, 34, 67, 53 }. Let's perform the insertion sort to sort in ascending order. During the sorting, what would the sequence look like after the second insertion (noting that before the first insertion, there is already one element sorted)?

- A. 63, 52, 37, 45, 71, 58, 42, 89, 34, 67, 53
- B. 52, 63, 37, 45, 71, 58, 42, 89, 34, 67, 53
- C. 52, 37, 63, 45, 71, 58, 42, 89, 34, 67, 53
- D. 37, 52, 63, 45, 71, 58, 42, 89, 34, 67, 53

答案正确：3 分　创建提问

R2-17　分数 3　　作者 陈越　单位 浙江大学

Given a binary tree with in-order traversal sequence { 4, 6, 1, 2, 5, 7, 3, 8 } and pre-order traversal sequence { 1, 6, 4, 3, 5, 2, 7, 8 }. What is the index of 2 if we number the nodes in level-order? (Obviously, the index of the root 1 is 1.)

- A. 7
- B. 3
- C. 5
- D. 2

答案正确：3 分　创建提问

R2-18　分数 3　　作者 杨子祺　单位 浙江大学

概览

答卷

提交列表

排名

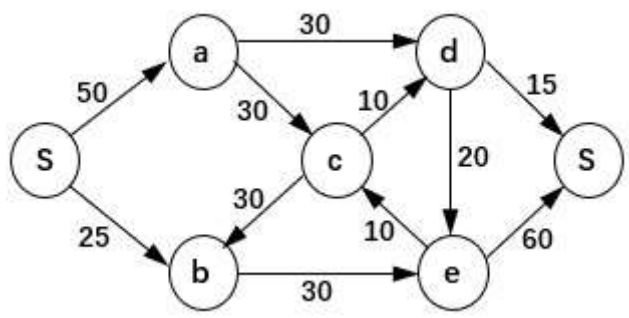
Given an integer sequence { 25, 84, 21, 47, 15, 27, 68, 35, 20 }. Let's perform shell sort in ascending order, using Shell's increment sequence by an increment 4 in the first run. After the second run, the integer sequence is

- A. 25, 15, 20, 47, 21, 35, 27, 84, 68
- B. 15, 27, 20, 35, 21, 47, 25, 84, 68
- C. 15, 21, 20, 25, 84, 27, 68, 35, 47
- D. 25, 15, 20, 27, 35, 21, 68, 84, 47

答案正确：3 分　创建提问

R2-19　分数 3　　作者 陈越　单位 浙江大学

The maximum flow in the network of the given figure is:



- A. 65
- B. 70
- C. 50
- D. 75

答案正确：3 分　创建提问

R2-20　分数 3　　作者 陈越　单位 浙江大学

The nodes in a complete binary tree are indexed in level-order (top-down, left-to-right) from 1. In a complete binary tree with 500 nodes, the node with the index 249

- A. has a left child, but no right child
- B. has no child
- C. has a right child, but no left child
- D. has two children

答案正确：3 分　创建提问

R5-1 Prim's algorithm 分数 6

作者 陈翔 单位 浙江大学

In computer science, Prim's algorithm is a greedy algorithm that finds a minimum spanning tree for a weighted undirected graph.

Please complete the following function `prim` to implement this algorithm. We assume all the edge weights are positive and store the graph as a weighted adjacency matrix `weight[][]`. Note that `weight[i][j] == 0` when no edge exists between vertex `i` and vertex `j`. Moreover, if `weight[i][j] > 0`, then it is the weight of the edge between these two vertices.

```
/*
 * Function: prim
 * --------------
 *   Find a minimum spanning tree for the given undirected
 *   graph by using Prim's algorithm
 *
 *   weight: the weighted adjacency matrix
 *   n: the number of vertices
 *
 *   returns: the total edge weights of the MST
 */
int prim(int weight[MAX][MAX], int n)
{
  int dist[MAX]; // distance from vertex i to the known part
  int prev[MAX]; // for tracing the edges of MST
  int known[MAX]; // 1 if the vertex i is checked, 0 if not

  // initialization
  for (int i = 0; i < n; i++)
  {
    dist[i] = INFINITY;
    prev[i] = -1;
    known[i] = 0;
  }

  dist[0] = 0; // start from vertex 0
  for (int k = 0; k < n; ++k)
  {
    // choose the vertex closest to the known part
    int min_dist = INFINITY;
    int min_vert = -1;
    for (int i = 0; i < n; i++)
    {
      if (!known[i] && dist[i] < min_dist)
      {
        min_dist = dist[i];
        min_vert = i;
      }
    }

    // relaxation of vertices adjacent to the chosen one
    known[min_vert] = 1;
    for (int i = 0; i < n; i++)
    {
      if (!known[i])
      {
        if ( weight[min_vert][i] > 0 && weight[min_vert][i] < dist[i]   3 分 )
        {
          dist[i] = weight[min_vert][i]   3 分 ;
          prev[i] = min_vert;
        }
      }
    }
  }

  // total edge weights
  int mst = 0;
  for (int i = 1; i < n; ++i)
    mst += dist[i];
  return mst;
}
```

答案正确：6 分 创建提问

R5-2 Quadratic Hashing 分数 6

作者 朱建科 单位 浙江大学

Find Key from a hash table H with quadratic probing. The function **Find** should return the position of the Key in hash table. If Key value is not in hash table, the function should return -1.(When testing, the values of elements of array TheCells in Hashtbl will be initialized to -1, which represents empty.)

```
int Hash(const int Key,int TableSize);

typedef int ElementType;
typedef unsigned int Position;

struct HashTbl{
   int TableSize;
   ElementType *TheCells;
};
typedef struct HashTbl *HashTable;

Position Find(ElementType Key, HashTable H){
   Position CurrentPos = Hash(Key, H->TableSize);
   int CollisionNum = 0;
   while( H->TheCells[CurrentPos] != Key    3 分 ){
      if(H->TheCells[CurrentPos] == -1)
         return -1;
      CurrentPos += 2 * (++CollisionNum) - 1    3 分 ;
      if(CurrentPos >= H->TableSize)
         CurrentPos -= H->TableSize;
   }
   return CurrentPos;
}
```

答案正确：6 分 创建提问

题目描述

R6-1 A+B with Binary Search Trees 分数 8

作者 郑友怡 单位 浙江大学

A Binary Search Tree (BST) is recursively defined as a binary tree which has the following properties:

- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than or equal to the node's key.
- Both the left and right subtrees must also be binary search trees.

Given two binary search trees, T1 and T2, and an integer N, you are supposed to find a number A from T1 and a number B from T2 such that A+B=N.

Structure description:

The tree structure is shown below:

```
typedef struct TreeNode {
    int val;
    struct TreeNode *left, *right;
} TreeNode;
```

Function definition:

You should find all solutions satisfying A+B=N. The function definition is as follows:

```
int findSum(TreeNode *root1, TreeNode *root2, int target, int a[], int b[]);
```

`root1` and `root2` are the roots for T1 and T2 respectively. The number of nodes for each tree $n \le 2 \times 10^4$,

and key value of each node k: $-2 \times 10^9 \le k \le 2 \times 10^9$. `N` is the target number. For each test case, the function returns `nsum`, the number of solutions, then saves all the solutions in `a` and `b`. That is, each solution is in the format `N = a[i] + b[i]` $(0 \le i < nsum)$. In case there are multiple solutions, values in `a` must be in ascending order. Note: the same solution should be saved *only once*.

Input Specification:

Each input file contains one test case. Each case gives the information of T1, T2, and N, in the following format:
The first line contains a positive integer n_1 ($\le 2 \times 10^4$), which is the number of nodes in T1. Then n_1 lines follow, where the i-th line contains the key value k ($-2 \times 10^9 \le k \le 2 \times 10^9$) and the parent node index of the i-th node ($0 \le i < n_1$). Since the root has no parent, its parent index is defined to be -1. After T1, T2 is given in the same format as T1. Finally, the last line gives the target N (with the same range of k).

Output Specification:

Firstly, you should print in a line `true` if at least one solution exists, then output all the solutions in the following lines, each in the format

`N = A + B`. In the case of multiple solutions, you must ensure that values `A` are in ascending order. Note: the same equation should be printed *only once*. If there is no solution, simply print `false`.

Test program example:

```c
#include <stdio.h>
#include <stdlib.h>

#define MAX_SIZE 20000

int values_1[MAX_SIZE];
int parents_1[MAX_SIZE];

int values_2[MAX_SIZE];
int parents_2[MAX_SIZE];

typedef struct TreeNode {
    int val;
    struct TreeNode *left, *right;
} TreeNode;

TreeNode *buildTree(int values[], int parents[], int n); /* details omitted */

int findSum(TreeNode *root1, TreeNode *root2, int N, int a[], int b[]);

int main() {

    int n;
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        scanf("%d %d", &values_1[i], &parents_1[i]);
    }

    int m;
    scanf("%d", &m);
    for (int i = 0; i < m; i++) {
        scanf("%d %d", &values_2[i], &parents_2[i]);
    }

    int N;
    scanf("%d", &N);

    TreeNode *root1 = buildTree(values_1, parents_1, n);
    TreeNode *root2 = buildTree(values_2, parents_2, m);

    int a[MAX_SIZE];
    int b[MAX_SIZE];
    int nsum = findSum(root1, root2, N, a, b);

    if (nsum) {
        printf("true\n");
        for (int i = 0; i < nsum; i++)
            printf("%d = %d + %d\n", N, a[i], b[i]);
    } else
        printf("false\n");

    return 0;
}

int findSum(TreeNode *root1, TreeNode *root2, int N, int a[], int b[]) {
    /* your implementation of this function */
}
```

Sample Input 1:

```
8
12 2
16 5
13 4
18 5
15 -1
17 4
14 2
18 3
7
20 -1
```

```
16 0
25 0
13 1
18 1
21 2
28 2
36
```

Sample Output 1:

```
true
36 = 15 + 21
36 = 16 + 20
36 = 18 + 18
```

Sample Input 2:

```
5
10 -1
5 0
15 0
2 1
7 1
3
15 -1
10 0
20 0
40
```

Sample Output 2:

```
false
```

代码长度限制	16 KB
时间限制	100 ms
内存限制	64 MB